

The Schur degree of block sums: $L(4) = 16$ and $L(5) \geq 49$

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Abstract

Eliahou and Revuelta defined a number $L(n)$ through the Schur degree of the set of block sums of a sequence of positive integers. They proved $S(n-1) + 1 \leq L(n) \leq R_{n-1}(3) - 1$ and $S(n) \leq nL(n)$, and they conjectured $L(n) = S(n-1) + 1$, which would give the recursive bound $S(n) \leq n(S(n-1) + 1)$ for the Schur numbers and $S(6) \leq 966$. For $n = 4$ they proved $14 \leq L(4) \leq 16$ and left the value open. We prove $L(4) = 16$ with one sequence of length 15 and an explicit partition of its 25 block sums into three sumfree sets. We then give a lift that turns a partition of the nonzero elements of $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2}$ into $n-1$ sets that are sumfree in the group into the lower bound $L(n) \geq m_1 m_2$. A partition of $\mathbb{Z}_7 \times \mathbb{Z}_7$ gives $L(5) \geq 49$. So the conjecture is false for $n = 4$ and $n = 5$. The two recursive consequences, Conjectures 5.7 and 5.8, remain open. The value $L(4) = 16$, the bound $L(5) \geq 49$ and the lift are also formalized in Lean 4 with Mathlib.

1. Introduction

A set S of integers is *sumfree* when there are no $x, y, z \in S$ with $x + y = z$; the elements x and y may be equal. The Schur number $S(n)$ is the largest N such that $\{1, \dots, N\}$ is the union of n sumfree sets. Only $S(1), \dots, S(5) = 1, 4, 13, 44, 160$ are known; $S(4) = 44$ is due to Baumert [Bau61] and $S(5) = 160$ to Heule [Heu18]. For $S(6)$, Eliahou and Revuelta give the range $536 \leq S(6) \leq 1836$ [ER21]; the lower bound is due to Fredricksen and Sweet [FS00].

Eliahou and Revuelta [ER21] introduced the *Schur degree* $\text{sdeg}(X)$ of a set X : the least number of sumfree sets that cover X . They applied it to the set $\hat{A}$ of block sums of a finite sequence A of positive integers, and they defined $L(n)$ as the least length L such that every sequence of L positive integers with average at most n has $\text{sdeg}(\hat{A}) \geq n$ (Definition 2.1 below). They proved

$$S(n-1) + 1 \leq L(n) \leq R_{n-1}(3) - 1 \quad \text{and} \quad S(n) \leq nL(n),$$

where $R_k(3)$ is the Ramsey number of the triangle in k colours, and they stated three conjectures:

- Conjecture 5.6: $L(n) = S(n-1) + 1$ for $n \geq 2$;
- Conjecture 5.7: $S(n) \leq n(S(n-1) + 1)$;
- Conjecture 5.8: $S(6) \leq 966$.

Conjecture 5.6 implies the other two. The two bounds on $L(n)$ agree for $n = 2$ and $n = 3$, which gives $L(2) = 2$ and $L(3) = 5$. For $n = 4$ they give $14 \leq L(4) \leq 16$. Eliahou and Revuelta conjectured $L(4) = 14$, asked whether every sequence of length 14 with average at most 4 has $\text{sdeg}(\hat{A}) \geq 4$, and wrote “We do not know yet”.

We prove the following.

Theorem 1.1. $L(4) = 16$.

Theorem 1.2. $L(5) \geq 49$. So $49 \leq L(5) \leq 61$.

Conjecture 5.6 predicts $L(4) = 14$ and $L(5) = 45$, so it is false for $n = 4$ and $n = 5$. At $n = 4$ the value of $L(n)$ is the upper bound $R_{n-1}(3) - 1$, not the lower bound. Theorem 1.1 answers the question of [ER21] in the negative: the first 14 entries of the sequence W of Section 3 have average $15/7$ and Schur degree of the block sums at most 3.

The proof of Theorem 1.1 is short and can be checked by hand. Theorem 1.2 comes from a general lift (Lemma 4.1 and Corollary 4.2) from partitions of the nonzero elements of $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2}$ into sets that are sumfree in the group.

Conjectures 5.7 and 5.8 remain open. Conjecture 5.7 holds for $n = 4$ and $n = 5$ by the known values ($44 \leq 4 \cdot 14$, $160 \leq 5 \cdot 45$). Conjecture 5.8 follows from Conjecture 5.6 at $n = 6$, that is, from $L(6) = 161$, and our results do not decide that case (Section 6).

Theorem 1.1, the bound $L(5) \geq 49$, Lemma 4.1 and Corollary 4.2 are also formalized in Lean 4 with Mathlib (Section 7). The upper bound $L(5) \leq 61$ uses $R_4(3) \leq 62$ [FKR04], which is not formalized; the formal upper bound is $L(5) \leq 65$.

2. Definitions and results of Eliahou and Revuelta

All sets below are sets of positive integers, except in the groups of Sections 4 to 6. A set X is *covered by n sumfree sets* when it is a subset of the union of n sumfree sets. The *Schur degree* $\text{sdeg}(X)$ is the least $n \geq 1$ such that n sumfree sets cover X [ER21, Definition 2.1]. For example, $\text{sdeg}(\{1, \dots, N\}) \leq n$ holds for $N \leq S(n)$ and fails for $N > S(n)$.

Let $A = (a_1, \dots, a_L)$ be a sequence of positive integers, of length $|A| = L$. A *block* of A is a run $(a_i, \dots, a_j)$ of consecutive entries with $1 \leq i \leq j \leq L$, and $\hat{A}$ is the set of the sums of the blocks. The *average* of A is $\mu(A) = (a_1 + \dots + a_L)/L$.

Definition 2.1 [ER21, Definition 5.1]. Let $n \geq 2$. Then $L(n)$ is the smallest positive integer with the following property: every sequence A of positive integers with $|A| = L(n)$ and $\mu(A) \leq n$ has $\text{sdeg}(\hat{A}) \geq n$.

For $n \geq 2$, the inequality $\text{sdeg}(\hat{A}) \geq n$ holds when no $n - 1$ sumfree sets cover $\hat{A}$, and it fails when some $n - 1$ sumfree sets cover $\hat{A}$. The property is about sequences of one length. A sequence of length $L + 1$ and average at most n need not contain L consecutive entries of average at most n , so a lower bound $L(n) \geq m$ needs a counterexample at every length L with $1 \leq L < m$.

We use the following results of [ER21]; the preprint [ER20] has the same numbering.

- **Proposition 2.5.** If B is a block of A , then $\hat{B} \subseteq \hat{A}$.
- **Proposition 2.7.** Let $X = \{x_0 < x_1 < \dots < x_L\}$ be a finite set of integers, and let $A = \Delta X = (x_1 - x_0, \dots, x_L - x_{L-1})$ be its sequence of jumps. Then $\hat{A} = (X - X) \cap \mathbb{Z}_{>0}$.
- **Proposition 3.2.** If B is a block of A , then $\text{sdeg}(\hat{B}) \leq \text{sdeg}(\hat{A})$.
- **Theorem 4.1.** If $|A| \geq R_q(3) - 1$, then $\text{sdeg}(\hat{A}) \geq q + 1$.
- **Proposition 5.3.** $S(n - 1) + 1 \leq L(n) \leq R_{n-1}(3) - 1$ for $n \geq 2$.
- **Theorem 5.4.** $S(n) \leq n L(n)$.

3. The value $L(4) = 16$

Let

$$W = (1, 1, 1, 6, 1, 1, 1, 7, 1, 1, 1, 6, 1, 1, 1).$$

Its length is 15 and its sum is 31. Its prefix sums form the set

$$X = \{0, 1, 2, 3\} + \{0, 9, 19, 28\} = \{0, 1, 2, 3, 9, 10, 11, 12, 19, 20, 21, 22, 28, 29, 30, 31\},$$

and $W = \Delta X$.

Lemma 3.1. $\hat{W} = [1, 3] \cup [6, 13] \cup [16, 22] \cup [25, 31]$. This set has 25 elements.

Proof. By Proposition 2.7, $\hat{W}$ is the set of positive differences of elements of X . Write each element of X as $u + t$ with $u \in \{0, 1, 2, 3\}$ and $t \in T = \{0, 9, 19, 28\}$. Two elements with the same t give the differences 1, 2 and 3. The positive differences of T are 9, 10, 19 and 28. A positive difference $t - t'$ plus $u - u' \in [-3, 3]$ gives the intervals $[6, 12]$, $[7, 13]$, $[16, 22]$ and $[25, 31]$, and every value in these intervals occurs. $\square$

Lemma 3.2. The three sets

$$\begin{aligned} C_0 &= \{1, 3, 8, 12, 18, 22, 28\}, \\ C_1 &= \{2, 6, 7, 10, 11, 25, 26, 29, 30\}, \\ C_2 &= \{9, 13, 16, 17, 19, 20, 21, 27, 31\} \end{aligned}$$

are sumfree, and they partition $\hat{W}$. So $\text{sdeg}(\hat{W}) \leq 3$.

Proof. The sets are disjoint, they have $7 + 9 + 9 = 25$ elements, and they lie in $\hat{W}$; by Lemma 3.1 they partition $\hat{W}$. The table lists, for each set, every sum $x + y$ with $x \leq y$ in the set (including $x = y$) and $x + y \leq 31$. No listed sum is in the set. Sums above 31 are not in $\hat{W}$.

set	sums $x + y \leq 31$ with $x \leq y$
C_0	2, 4, 6, 9, 11, 13, 15, 16, 19, 20, 21, 23, 24, 25, 26, 29, 30, 31
C_1	4, 8, 9, 12, 13, 14, 16, 17, 18, 20, 21, 22, 27, 28, 31
C_2	18, 22, 25, 26, 28, 29, 30

$\square$

Proof of Theorem 1.1. By Proposition 5.3 and $R_3(3) = 17$ [GG55], $L(4) \leq 16$. (Directly: every sequence of length $16 = R_3(3) - 1$ has $\text{sdeg}(\hat{A}) \geq 4$ by Theorem 4.1.)

For the lower bound, let $1 \leq L \leq 15$ and let B be the block of the first L entries of W . The sums of these blocks, for $L = 1, \dots, 15$, are

$$1, 2, 3, 9, 10, 11, 12, 19, 20, 21, 22, 28, 29, 30, 31.$$

Each is at most $4L$, so $\mu(B) \leq 4$. By Proposition 3.2 and Lemma 3.2, $\text{sdeg}(\hat{B}) \leq \text{sdeg}(\hat{W}) \leq 3 < 4$. So the property of Definition 2.1 for $n = 4$ fails at every length $L \leq 15$, and $L(4) = 16$. $\square$

Remark 3.3 (least sums; computer check). A SAT computation shows that every sequence A of length 14 with $\text{sdeg}(\hat{A}) \leq 3$ has sum at least 30. The solver CaDiCaL [BFF24] reported that the formula which allows every sum at most 29 is unsatisfiable, and drat-trim [WHH14] verified the proof of unsatisfiability. An independent exhaustive search without a SAT solver agrees. The first 14 entries of W have sum 30. By Proposition 3.2 the least sum at length 15 is then 31, which W attains. An exhaustive search found that W is the only sequence A of length 15 with sum at most 31 and $\text{sdeg}(\hat{A}) \leq 3$.

Remark 3.4 (a Ramsey colouring; computer check). The construction in the proof of [ER21, Theorem 4.1] turns Lemma 3.2 into a 3-colouring of the edges of the complete graph on X : the edge $\{x, y\}$ gets the index of the set that contains $|x - y|$. This colouring of K_{16} has no monochromatic triangle. A computer check shows that each colour class is a strongly regular graph with parameters $(16, 5, 0, 2)$, isomorphic to the Clebsch graph (the folded 5-cube).

4. A lift from finite abelian groups

Let G be an abelian group. A set $C \subseteq G$ is *sumfree in G* when there are no $x, y, z \in C$ with $x + y = z$ in G ($x = y$ allowed).

Lemma 4.1. Let $m_1, m_2 \geq 1$ and $q \geq 1$, and let $G = \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2}$. Suppose that the sets $C_1, \dots, C_q$ are sumfree in G and that their union contains $G \setminus \{0\}$. Let $M \geq 3m_1 - 2$, let

$$X = \{u + Mj : 0 \leq u \leq m_1 - 1, 0 \leq j \leq m_2 - 1\},$$

and let $A = \Delta X$. Then $|A| = m_1 m_2 - 1$, every entry of A is positive, $\text{sdeg}(\hat{A}) \leq q$, and for $0 \leq L \leq m_1 m_2 - 1$ the sum of the first L entries of A is

$$x_L = (L \bmod m_1) + M \lfloor L/m_1 \rfloor.$$

Proof. Since $M \geq m_1$, the $m_1 m_2$ numbers $u + Mj$ are distinct. In increasing order they are $x_L = u + Mj$ with $L = jm_1 + u$ and $0 \leq u \leq m_1 - 1$. So $|A| = m_1 m_2 - 1$, the entries of A are positive, and the sum of the first L entries is $x_L - x_0 = x_L$.

By Proposition 2.7, $\hat{A} = (X - X) \cap \mathbb{Z}_{>0}$. Let $d \in \hat{A}$, $d = (u + Mj) - (u' + Mj')$, and put $r = u - u'$ and $e = j - j'$. Then $d = r + Me$ with $|r| \leq m_1 - 1$ and $|e| \leq m_2 - 1$. If $e < 0$, then $d \leq (m_1 - 1) - M < 0$, which is false. So $e \geq 0$, and if $e = 0$ then $r = d \geq 1$.

The pair (r, e) depends only on d . Indeed, if $r + Me = r' + Me'$ with $|r|, |r'| \leq m_1 - 1$, then $M|e - e'| = |r' - r| \leq 2(m_1 - 1) < M$, so $e = e'$ and $r = r'$. Put

$$\pi(d) = (r \bmod m_1, e \bmod m_2) \in G.$$

Then $\pi(d) \neq 0$: we have $0 \leq e \leq m_2 - 1$, and $e = 0$ gives $1 \leq r \leq m_1 - 1$. Give d an index i with $\pi(d) \in C_i$.

Suppose that $d_1 + d_2 = d_3$ with $d_1, d_2, d_3 \in \hat{A}$ ($d_1 = d_2$ allowed). Then $r_1 + r_2 - r_3 = M(e_3 - e_1 - e_2)$. The left side has absolute value at most $3(m_1 - 1) < M$, so both sides are 0. So $\pi(d_1) + \pi(d_2) = \pi(d_3)$ in G , and d_1, d_2, d_3 do not all have the same index, because each C_i is sumfree in G . So the q index classes are sumfree sets that cover $\hat{A}$, and $\text{sdeg}(\hat{A}) \leq q$. $\square$

The hypothesis $q \geq 1$ is needed only in one degenerate case: for $m_1 = m_2 = 1$ the sequence A is empty, and $\text{sdeg}(\emptyset) = 1$. The proof uses $M \geq 3m_1 - 2$. For smaller M the colouring of the proof often fails in random tests, but the conclusion can still hold: for $m_1 = m_2 = 7$ and $M = 16$ a SAT solver found a cover of $\hat{A}$ by four sumfree sets.

Corollary 4.2. Let $n \geq 3$ and $m_1, m_2 \geq 1$. If $n - 1$ sets that are sumfree in $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2}$ cover its nonzero elements, then $L(n) \geq m_1 m_2$.

Proof. Take $M = 3m_1 - 2$ in Lemma 4.1. For $1 \leq L \leq m_1 m_2 - 1$, write $L = jm_1 + u$ with $0 \leq u \leq m_1 - 1$. Then $x_L = u + (3m_1 - 2)j < 3(u + m_1 j) = 3L$ when $L \geq 1$. So the first L entries of A form a block B with $\mu(B) < 3 \leq n$, and $\text{sdeg}(\hat{B}) \leq \text{sdeg}(\hat{A}) \leq n - 1$ by Proposition 3.2. So the property of Definition 2.1 fails at every length $L \leq m_1 m_2 - 1$, and $L(n) \geq m_1 m_2$. $\square$

Remarks.

1. For $m_2 = 1$ the corollary gives nothing new. If $\mathbb{Z}_N \setminus \{0\}$ is covered by $n - 1$ sets sumfree in $\mathbb{Z}_N$, then their restriction to $[1, N - 1]$ is a cover by $n - 1$ sumfree sets of integers, so $N - 1 \leq S(n - 1)$ and the bound $L(n) \geq N$ is at most the lower bound of Proposition 5.3. The gain comes from non-cyclic groups.
2. A partition of $(\mathbb{Z}_4 \times \mathbb{Z}_4) \setminus \{0\}$ into three sets that are sumfree in the group exists [WSW72], as cited in [Ana23]. With it, Corollary 4.2 gives $L(4) \geq 16$ again.
3. A cover of $G \setminus \{0\}$ by q sets that are sumfree in G gives a q -colouring of the edges of the complete graph on $|G|$ vertices with no monochromatic triangle. For symmetric sets the edge $\{g, h\}$ gets the colour of $g - h$ [GG55]; for sets that are not symmetric one uses an ordering of G [AH72], as cited in [BCR25]. So $|G| < R_q(3)$, and Corollary 4.2 never gives more than the upper bound of Proposition 5.3.

5. The bound $L(5) \geq 49$

Proof of Theorem 1.2. The four sets in the table partition $(\mathbb{Z}_7 \times \mathbb{Z}_7) \setminus \{0\}$. Each has 12 elements, is closed under negation, and is sumfree in the group; this is a finite check of $4 \cdot 12^2$ sums.

set	elements $(a, b) \in \mathbb{Z}_7 \times \mathbb{Z}_7$
C_0	(0,2) (0,5) (1,4) (2,3) (2,4) (3,2) (3,3) (4,4) (4,5) (5,3) (5,4) (6,3)
C_1	(1,1) (1,3) (1,5) (1,6) (3,0) (3,5) (4,0) (4,2) (6,1) (6,2) (6,4) (6,6)
C_2	(0,3) (0,4) (1,0) (1,2) (2,1) (2,6) (3,4) (4,3) (5,1) (5,6) (6,0) (6,5)
C_3	(0,1) (0,6) (2,0) (2,2) (2,5) (3,1) (3,6) (4,1) (4,6) (5,0) (5,2) (5,5)

Corollary 4.2 with $n = 5$ and $m_1 = m_2 = 7$ gives $L(5) \geq 49$. The upper bound $L(5) \leq R_4(3) - 1 \leq 61$ follows from Proposition 5.3 and $R_4(3) \leq 62$ [FKR04, Rad26]. $\square$

With $M = 19$ the sequence of Lemma 4.1 is

$$A = (1^6, 13, 1^6, 13, 1^6, 13, 1^6, 13, 1^6, 13, 1^6, 13, 1^6),$$

where 1^6 stands for six entries equal to 1. It has length 48, sum 120, and $|\hat{A}| = 84$; the largest average of a prefix is $19/7$. A direct computer check, with no solver, confirms that the colouring of the proof of Lemma 4.1 has no monochromatic solution of $x + y = z$ in $\hat{A}$.

A partition of $(\mathbb{Z}_7 \times \mathbb{Z}_7) \setminus \{0\}$ of this kind gives $R_4(3) \geq 50$ by Remark 3 of Section 4, a bound that Anabanti [Ana17] credits to Whitehead [Whi73]. The best known lower bound is $R_4(3) \geq 51$ [Chu73, Rad26]. Our partition was found by a SAT solver.

The limits of the lift at $n = 5$ (computer check). Corollary 4.2 gives no bound above $L(5) \geq 49$:

- a cyclic group $\mathbb{Z}_N$ with $N \geq 46$ has no cover of its nonzero elements by 4 sets sumfree in the group, because the restriction to $[1, N - 1]$ would give $S(4) \geq 45$ (Remark 1 of Section 4);
- the non-cyclic abelian groups of order 50 to 60 have no such cover. There are seven such groups (of orders 50, 52, 54, 56 and 60). For each, CaDiCaL reported the formula unsatisfiable, and drat-trim verified the proof;
- a group $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2}$ of order at least 62 has no such cover, because Corollary 4.2 would give $L(5) \geq 62 > 61$; and 61 is prime.

Anabanti [Ana23] excludes partitions into four symmetric sets for all groups of order 51 to 61; the computation above does not assume symmetry. We know no sequence of length 49 or more that shows $L(5) > 49$, and the exact value of $L(5)$ is open.

6. The case $n = 6$

For $n = 6$, Proposition 5.3 and $R_5(3) \leq 307$ [Rad26] give $161 \leq L(6) \leq 306$. Conjecture 5.6 at $n = 6$, that is $L(6) = 161$, is the case that would give Conjecture 5.8.

By Corollary 4.2, a cover of the nonzero elements of a group $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2}$ of order at least 162 by five sets that are sumfree in the group would refute Conjecture 5.6 at $n = 6$. By Remark 3 of Section 4 it would also give $R_5(3) \geq m_1 m_2 + 1 \geq 163$, which is above the best known lower bound $R_5(3) \geq 162$ [Exo94, Rad26]. A cyclic group of order at least 162 has no such cover, because $S(5) = 160$. So a refutation at $n = 6$ through Corollary 4.2 needs a non-cyclic group of order 162 to 306 with such a cover, and it would improve the lower bound for $R_5(3)$. Our solver searches did not find one; they were not exhaustive.

7. Formal verification

The results of Sections 3 to 5 are formalized in Lean 4 [MU21] with Mathlib [mC20] (Lean v4.33.1). The definitions follow [ER21], with sets of natural numbers and sequences given as lists of natural numbers; for a subset of the positive integers the Schur degree is the same in $\mathbb{N}$ and in $\mathbb{Z}$. The formal statements are:

- $L(4) = 16$;
- $49 \leq L(5) \leq 65$;
- Lemma 4.1 and Corollary 4.2, with covers that need not be partitions;
- Theorem 4.1 of [ER21] for sequences of natural numbers, and the upper bound $L(k+1) \leq \rho(k) - 1$ of Proposition 5.3, where $\rho(0) = 2$ and $\rho(k+1) = (k+1)(\rho(k) - 1) + 2$.

The bound $\rho(k)$ is the pigeonhole upper bound for $R_k(3)$, proved in the formalization. Since $\rho(3) = 17$, the formal proof of $L(4) \leq 16$ does not use the Greenwood–Gleason colouring. Since $\rho(4) = 66 > 62$, the formal upper bound for $L(5)$ is 65, not 61. The finite checks (Lemma 3.2, the prefix sums of W , and the partition of Section 5) are done by the Lean kernel with `decide`, without native code. The only axioms used are `propext`, `Classical.choice` and `Quot.sound`. As a check of the definitions, a separate Lean proof (not part of the library) shows that they give $L(2) = 2$ and $L(3) = 5$, the values of [ER21].

The six main statements are also checked with the leanprover/comparator tool [LFRO]. The tool compares the statements, with every definition that they use, against a separate file that imports only Mathlib, and it checks the proofs with the Lean kernel and with the independent nanoda kernel.

Data and code availability. The Lean source, the comparator configuration and programs for the computer checks are in the repository <https://github.com/mysticflounder/schur-degree-block-sums>. The programs check the witnesses of Theorems 1.1 and 1.2 and the parameters of the colouring of Remark 3.4 without a solver, and they write the SAT formulas of Remark 3.3 and of Section 5; the repository gives the SHA-256 digests of the formulas of Section 5. The exhaustive searches of Remark 3.3, the isomorphism test of Remark 3.4, the Lean check of $L(2)$ and $L(3)$, and the solver searches of Section 6 were made with programs that are not in the repository.

8. Related work

We found no work after [ER21] on $L(n)$ or on Conjectures 5.6 to 5.8. The citation databases Crossref, OpenAlex, OpenCitations and Semantic Scholar list no work that cites [ER21] (September 2026). The textbook [Jun23] lists [ER21] among its references; we have not seen the text that cites it.

The use of partitions of finite groups into sumfree sets for Ramsey colourings is classical [GG55, AH72, Ana17, Ana23]. We found no earlier application of such partitions to block sums or to the Schur degree.

Acknowledgements and use of AI tools

{{NEEDS_ADAM_INPUT}} Draft statement, to be confirmed: The sequence W , the proofs, the computer checks and the Lean formalization were produced with the AI system Claude (Anthropic) under the direction of the author. Independent AI agents audited the proofs and the computations with their own code. The author has checked the content and takes responsibility for it.

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